

Source Code / Commands Used for Footprinting, Reconnaissance, and Network Scanning Project

1. Checking Internet Connectivity

Before starting reconnaissance, internet connectivity was verified on the Kali Linux machine.

```
ping google.com
```

If DNS resolution failed, connectivity was tested using Google DNS.

```
ping 8.8.8.8
```

2. Google Dorking

Google Dorks were used to identify publicly available sensitive information related to the target domain.

Finding Login Pages

```
site:example.com login
```

Finding Public PDF Files

```
site:example.com filetype:pdf
```

Finding Open Directories

```
intitle:"index of"
```

Finding Exposed SQL Files

```
site:example.com ext:sql
```

These searches helped identify publicly accessible resources and possible information leakage.

3. Whois Lookup

Whois lookup was performed to gather domain registration and ownership information.

```
whois example.com
```

The Whois query provided:

- Registrar information
 - Domain creation and expiration dates
 - Name server details
 - Administrative and technical contact details
-

4. NsLookup

NsLookup was used to identify DNS records and IP addresses associated with the target domain.

```
nslookup example.com
```

The output displayed:

- Domain IP address
 - DNS server information
-

5. Reverse DNS Lookup

A reverse lookup was performed to identify domain names associated with the discovered IP address.

```
nslookup 93.184.216.34
```

This helped identify additional systems or associated domains hosted on the same infrastructure.

6. Dig Command

The Dig command was used for advanced DNS enumeration.

Basic Dig Query

```
dig example.com
```

MX Record Enumeration

```
dig MX example.com
```

Name Server Enumeration

```
dig NS example.com
```

These commands helped identify:

- Mail servers
 - DNS name servers
 - Additional DNS-related information
-

7. Installing Recon-ng

Recon-ng was installed for automated reconnaissance and subdomain enumeration.

```
sudo apt install recon-ng -y
```

8. Starting Recon-ng

```
recon-ng
```

9. Installing Recon-ng Module

Inside Recon-ng:

```
marketplace install recon/domains-hosts/bing_domain_web
```

10. Loading the Module

```
modules load recon/domains-hosts/bing_domain_web
```

11. Setting the Target Domain

```
options set SOURCE example.com
```

12. Running Recon-ng Enumeration

```
run
```

Recon-ng attempted to identify:

- Subdomains
 - Associated hosts
 - Publicly exposed systems
-

13. Installing Sublist3r (Alternative Enumeration Tool)

Since some Recon-ng modules may require API access, Sublist3r was used as an alternative subdomain enumeration tool.

```
sudo apt install sublist3r -y
```

14. Running Subdomain Enumeration

```
sublist3r -d example.com
```

This command searched for subdomains associated with the target domain.

15. Performing Basic Nmap Scan

Nmap was used to identify open ports on the target system.

```
nmap 93.184.216.34
```

The scan identified:

- Open ports
 - Accessible network services
-

16. Service Version Detection

```
nmap -sV 93.184.216.34
```

This scan detected:

- Service versions
 - Running applications
 - Potential outdated software
-

17. Aggressive Network Scan

```
nmap -A 93.184.216.34
```

The aggressive scan provided:

- Operating system information
 - Service banners
 - Traceroute information
 - Additional enumeration details
-

18. Vulnerability Scanning

Nmap vulnerability scripts were used to identify known weaknesses.

```
nmap --script vuln 93.184.216.34
```

This scan checked for:

- Misconfigurations
 - Vulnerable services
 - Weak security settings
-

19. Saving Scan Results

The scan output was saved into a report file.

```
nmap -sV -oN scan_results.txt 93.184.216.34
```

20. Summary of Activities

The project successfully demonstrated:

- Open-source intelligence gathering
- DNS enumeration
- Whois information collection
- Subdomain discovery
- Port scanning
- Service detection
- Vulnerability identification

The collected information was later analyzed to identify security risks and recommend mitigation strategies for improving the target organization's cybersecurity posture.